

香港教育大學  
The Education University of Hong Kong

中學教育學士課程  
Bachelor of Education Secondary Programme

---

## **The Database Development Project (IIT3008): 2021–2022**

### **1. Introduction**

The goal of the project is to provide a realistic experience in the conceptual design, logical design, implementation, operation, and maintenance of a relational database and associated STEM applications. You are required to work in a group of 2 persons to produce mobile apps linked with a publicly available relational database to manage a school application. The due date is scheduled on 26<sup>th</sup> April 2022. You need to present the development and hand in the documents and software as specified in Section 2.2 on the due date.

### **2. Project Description**

Ms Lee Ho, the Principal of ABC Secondary School, would like to design and implement a database with the daily operation of the school tuck shop and provide parents with the information of the purchasing habits for their children. Therefore, Ms Lee decided to develop three mobile apps (1) to allow students to order food, (2) to let the parents to view and manage their purchases, and (3) to allow the administrator and Ms Ho to view the database and maintain the daily operation.

#### **2.1 Requirements**

1. The database must maintain information on an unknown number of students and parents as well as unknown number of product items available for purchase. The database must also maintain information about what type the product item is: dry food, wet food and soft drink. Examples of dry food include chips, biscuits and candy. Examples of wet food include fish balls, shao mai and soup noodles (templates of pre-made barcodes). To simplify your database design, each product item has only one supplier. One supplier can supply a number of products. The database must maintain information on the suppliers.
2. Parents and students are allowed to add money into their accounts. However, students cannot purchase any product if the remaining balance of their respective card is not sufficient or the parent has put a ban on any purchase by the student.
3. Ms Lee wants to be able to track the purchasing history on each student and be able to prepare reports showing the purchasing history of each product.
4. On every transaction by the students, a receipt will be listed including all the purchased items and the total cost. The remaining balance in the database will be shown.
5. Parents can obtain a report detailed out all the purchasing activities of their children and optionally suspend any purchase by their children.
6. Ms Lee can also generate back-orders if the stock falls below a pre-defined level.
7. All the necessary forms and reports to maintain the database should be provided.

## 2.2 Deliverables

1. A technical report outlines the following:
  - Software and hardware requirement
  - Database design includes:
    - Entity-Relationship Diagrams
    - Database tables
    - Attributes and definitions
    - Table connections
  - Query specifications
  - Form specifications
  - Report specifications
  - The detailed distribution of work loads for each member in the group
  - A small reflection on the project
2. A user report should include:
  - Installation (the database system and mobile apps)
  - How to start the system
  - How to navigate through the apps
  - Examples of screen and data entry forms
  - Instruction about data entry
  - Advice on how to respond to error messages or conditions
  - Examples of data output screens and reports
3. A demonstration by the group.
4. To evaluate and present the database system, the project will be executable in mobile apps connected to a relational database. The technical report, the user report and the complete software should also be submitted to the provided Google drive with the declaration sheet of originality.

## 2.3 Evaluation

The evaluation is based on the demonstration, the reports and the software. The demonstration will take into account of the functionality, smoothness and explanation. The evaluation of reports will take into account the organization, clarity, completeness and delivery. The software will be examined on the specification and implementation to the technical competency and professional interface.

**Marks Allocation of the Project Components**

| Component     | Item             | Mark |
|---------------|------------------|------|
| Reports       | Technical Report | 10%  |
|               | User Report      | 10%  |
| Demonstration |                  | 10%  |
| Software      |                  | 20%  |

-- END --